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(54) **PRINT COMPONENT WITH MEMORY ARRAY USING INTERMITTENT CLOCK SIGNAL**

(71) Applicant: **HEWLETT-PACKARD DEVELOPMENT COMPANY, L.P.**, Spring, TX (US)

(72) Inventors: **James Michael Gardner**, Corvallis, OR (US); **Scott A. Linn**, Corvallis, OR (US); **Michael W. Cumbie**, Corvallis, OR (US)

(73) Assignee: **Hewlett-Packard Development Company, L.P.**, Spring, TX (US)

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(51) **Int. Cl.**
B41J 2/045 (2006.01)

(52) **U.S. Cl.**
CPC ***B41J 2/04573*** (2013.01); ***B41J 2/04541*** (2013.01); ***B41J 2/04543*** (2013.01)

(58) **Field of Classification Search**
CPC B41J 2/04543
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,026,402 A	5/1977	Byrd
4,872,028 A	10/1989	Lloyd

(Continued)

FOREIGN PATENT DOCUMENTS

CN	1727186	2/2006
CN	1922019 A	2/2007

(Continued)

OTHER PUBLICATIONS

"Four-Terminal Sensing." Wikipedia: The Free Encyclopedia. Wikimedia Foundation Inc., retrieved at https://en.wikipedia.org/wiki/Four-terminal_sensing, Sep. 16, 2018, 3 pages.

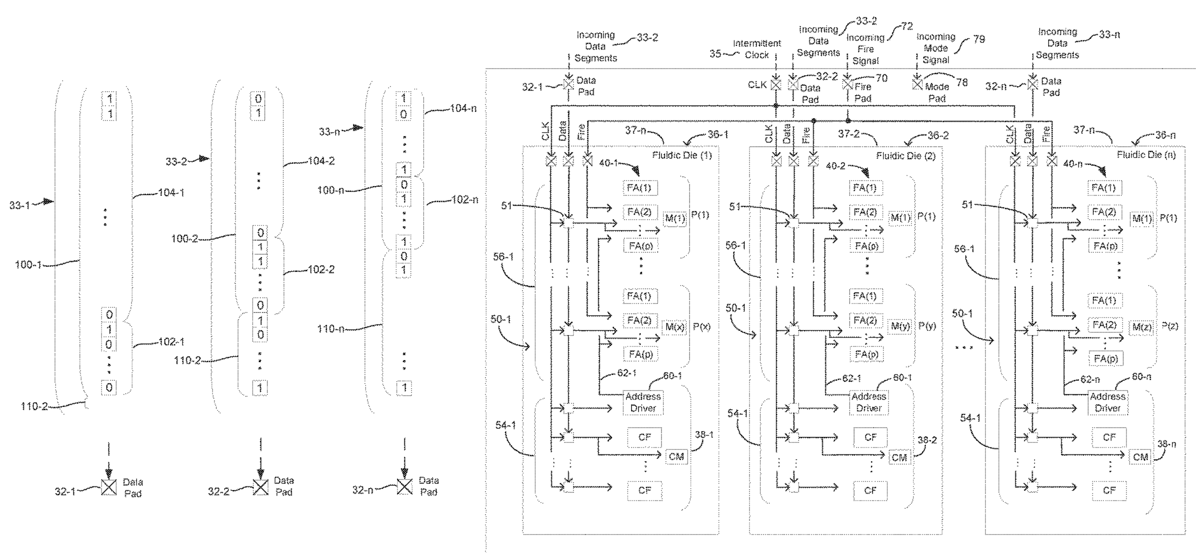
(Continued)

Primary Examiner — Shelby L Fidler
(74) *Attorney, Agent, or Firm* — Foley & Lardner LLP

(57) **ABSTRACT**

A print component includes a plurality of data pads, a clock pad to receive an intermittent clock signal, and a plurality of actuator groups each corresponding to a different liquid type and to a different one of the data pads. Each actuator group includes a plurality of configuration functions, an array of fluid actuators, and an array of memory elements including a first portion corresponding to the plurality of configuration functions and a second portion corresponding to the array of fluid actuators. Each time the intermittent clock signal is present on the clock pad, the array of memory elements to serially load a segment of data bits from the corresponding data pad, including loading a first portion of data bits into the first portion of memory elements, and loading a second portion of data bits into the second portion of memory elements.

27 Claims, 9 Drawing Sheets



(56) **References Cited**

U.S. PATENT DOCUMENTS

5,422,662	A	6/1995	Fukushima et al.	
5,541,629	A	7/1996	Saunders et al.	
5,751,302	A	5/1998	Rezanka	
5,917,509	A	6/1999	Becerra	
6,176,569	B1	1/2001	Anderson et al.	
6,302,507	B1	10/2001	Prakash et al.	
6,547,356	B2	4/2003	Edelen et al.	
6,616,256	B1	9/2003	Cook et al.	
6,698,862	B1	3/2004	Choi et al.	
7,104,624	B2	9/2006	Schloeman et al.	
7,159,959	B2	1/2007	Schremp	
7,252,353	B2	8/2007	Silverbrook et al.	
7,266,661	B2	9/2007	Walmsley	
7,267,417	B2	9/2007	Silverbrook et al.	
7,419,236	B2	9/2008	Masuda	
7,578,569	B2	8/2009	Silverbrook	
7,738,137	B2	6/2010	Graf et al.	
7,758,141	B2	7/2010	Sakurai	
7,832,842	B2	11/2010	Jackson Pulver et al.	
7,866,778	B2	1/2011	Silverbrook et al.	
7,976,115	B2	7/2011	Gibson et al.	
8,199,342	B2	6/2012	Martin	
8,348,373	B2	1/2013	Martin et al.	
8,353,567	B1	1/2013	Banerjee et al.	
8,556,364	B2	10/2013	Hoisington et al.	
8,777,364	B2	7/2014	Carrobe et al.	
8,864,260	B1	10/2014	Ge et al.	
9,415,585	B1	8/2016	Goyen et al.	
9,597,894	B2	3/2017	Buxton et al.	
9,776,395	B2	10/2017	Anderson et al.	
10,821,735	B2	11/2020	Anderson et al.	
11,364,719	B2 *	6/2022	Gardner	B41J 2/04541
11,407,218	B2	8/2022	Rossi et al.	
11,413,862	B2	8/2022	Linn et al.	
11,485,134	B2	11/2022	Rossi et al.	
11,559,985	B2	1/2023	Linn et al.	
2003/0063297	A1	4/2003	Dodd	
2004/0104973	A1	6/2004	Huang et al.	
2004/0183842	A1	9/2004	Kobayashi et al.	
2005/0110814	A1 *	5/2005	Imai	B41J 2/04588 347/5
2005/0190217	A1	9/2005	Wade et al.	
2005/0206944	A1	9/2005	Walmsley et al.	
2006/0050100	A1 *	3/2006	Kondoh	B41J 2/04581 347/19
2006/0092437	A1	5/2006	Martin	
2006/0109296	A1	5/2006	Shamoun et al.	
2006/0132518	A1	6/2006	Jackson Pulver et al.	
2006/0139380	A1	6/2006	Walmsley et al.	
2006/0181558	A1	8/2006	Walmsley et al.	
2007/0095928	A1	5/2007	Balinsky et al.	
2007/0126769	A1 *	6/2007	Usui	B41J 2/2128 347/15
2007/0177169	A1	8/2007	Han	
2007/0296743	A1 *	12/2007	Sakurai	B41J 2/04543 347/5
2008/0036804	A1	2/2008	Sakurai	
2008/0123641	A1	5/2008	Park	
2008/0238969	A1	10/2008	Kasai	
2008/0303868	A1 *	12/2008	Sakurai	B41J 2/14072 347/50
2009/0009545	A1 *	1/2009	Takei	B41J 2/0458 347/10
2009/0278879	A1 *	11/2009	Takeuchi	B41J 2/0458 347/12
2009/0278890	A1 *	11/2009	Yamaguchi	B41J 2/04541 347/59
2010/0124329	A1	5/2010	Lyman	
2010/0156992	A1 *	6/2010	Xie	B41J 2/155 347/49
2013/0063511	A1 *	3/2013	Gardner	B41J 2/04573 347/14
2013/0257949	A1	10/2013	Qiu et al.	
2014/0078223	A1	3/2014	Ohmura	
2014/0111570	A1	4/2014	Nitta et al.	
2014/0320558	A1	10/2014	Ge et al.	
2015/0124019	A1	5/2015	Cruz-Uribe et al.	
2015/0266324	A1	9/2015	Qu et al.	
2016/0089885	A1	3/2016	Edelen et al.	
2017/0239944	A1	8/2017	Martin et al.	
2017/0242633	A1	8/2017	Martin et al.	
2017/0272231	A1	9/2017	Chen	
2017/0320320	A1	11/2017	Martin et al.	
2018/0050537	A1	2/2018	Bakker et al.	
2018/0147839	A1	5/2018	Ge et al.	
2018/0236761	A1	8/2018	Taniguchi et al.	
2018/0236762	A1	8/2018	Negishi et al.	
2018/0288276	A1	10/2018	Sato et al.	
2019/0061347	A1	2/2019	Bakker et al.	
2021/0206164	A1	7/2021	Gardner et al.	
2021/0213731	A1	7/2021	Linn et al.	
2021/0221120	A1	7/2021	Linn et al.	
2021/0354453	A1	11/2021	Martin et al.	
2021/0362494	A1	11/2021	Gardner et al.	

FOREIGN PATENT DOCUMENTS

CN	101092076	A	12/2007
CN	101522428	A	9/2009
CN	101574865	A	11/2009
CN	101784392	A	7/2010
CN	103358701	A	10/2013
CN	107073940	A	8/2017
CN	107206816	A	9/2017
CO	5570691	A2	10/2005
CO	7160043	A2	1/2015
EP	0972374	A1	1/2000
EP	1212197	A1	6/2002
EP	3717254	B1	12/2023
ES	2788140	A2	10/2020
ES	2885775	T3	12/2021
ES	2892033	T3	2/2022
JP	59-208977	A	11/1984
JP	02-208052	A	8/1990
JP	2004282330		10/2004
JP	2004343468		12/2004
JP	2005-205770	A	8/2005
JP	2007-525342	A	9/2007
JP	2009-292148	A	12/2009
JP	2011075387		4/2011
JP	2016-032872	A	3/2016
JP	2016-165822	A	9/2016
JP	2018-505077	A	2/2018
KR	10-0739789	B1	7/2007
KR	10-2017-0109239	A	9/2017
TW	200533107	A	10/2005
TW	201346749	A	11/2013
TW	201601509	A	1/2016
WO	99/40702	A1	8/1999
WO	01/10647	A1	2/2001
WO	2016/089371	A1	6/2016
WO	2016/130157	A1	8/2016
WO	2017/180142	A1	10/2017
WO	2018/071034	A1	4/2018
WO	2018/080479	A1	5/2018
WO	2018/080480	A1	5/2018
WO	2018/136084	A1	7/2018
WO	2018/190855	A1	10/2018
WO	2019/009902	A1	1/2019
WO	2019/017951	A1	1/2019
WO	2019/195816	A1	10/2019

OTHER PUBLICATIONS

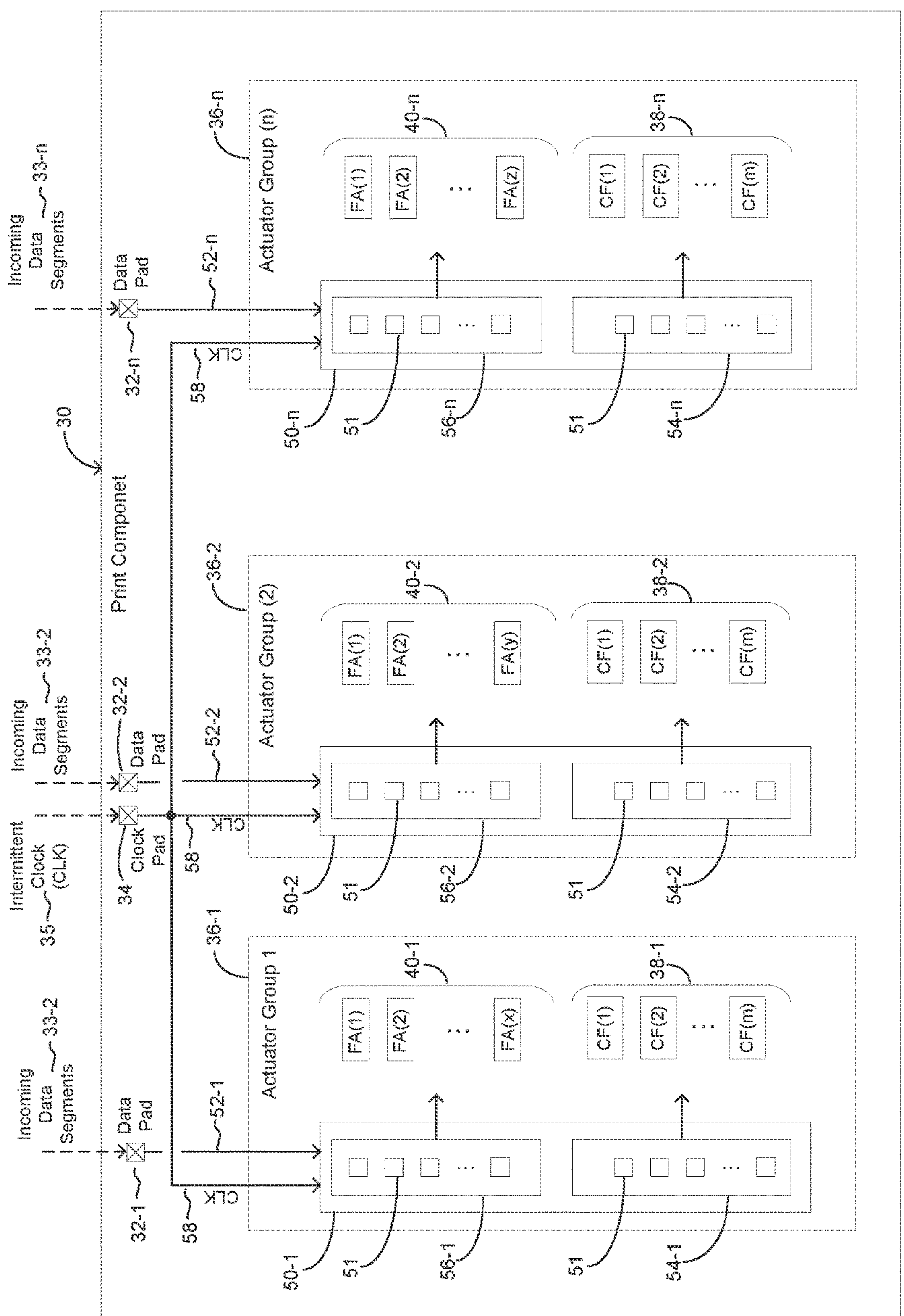
"What is a Kelvin connection and when should it be used?" retrieved at http://www.calibratorsinc.com/images/applications.pdf, Dated on or before Jan. 2019, 2 pages.

(56) **References Cited**

OTHER PUBLICATIONS

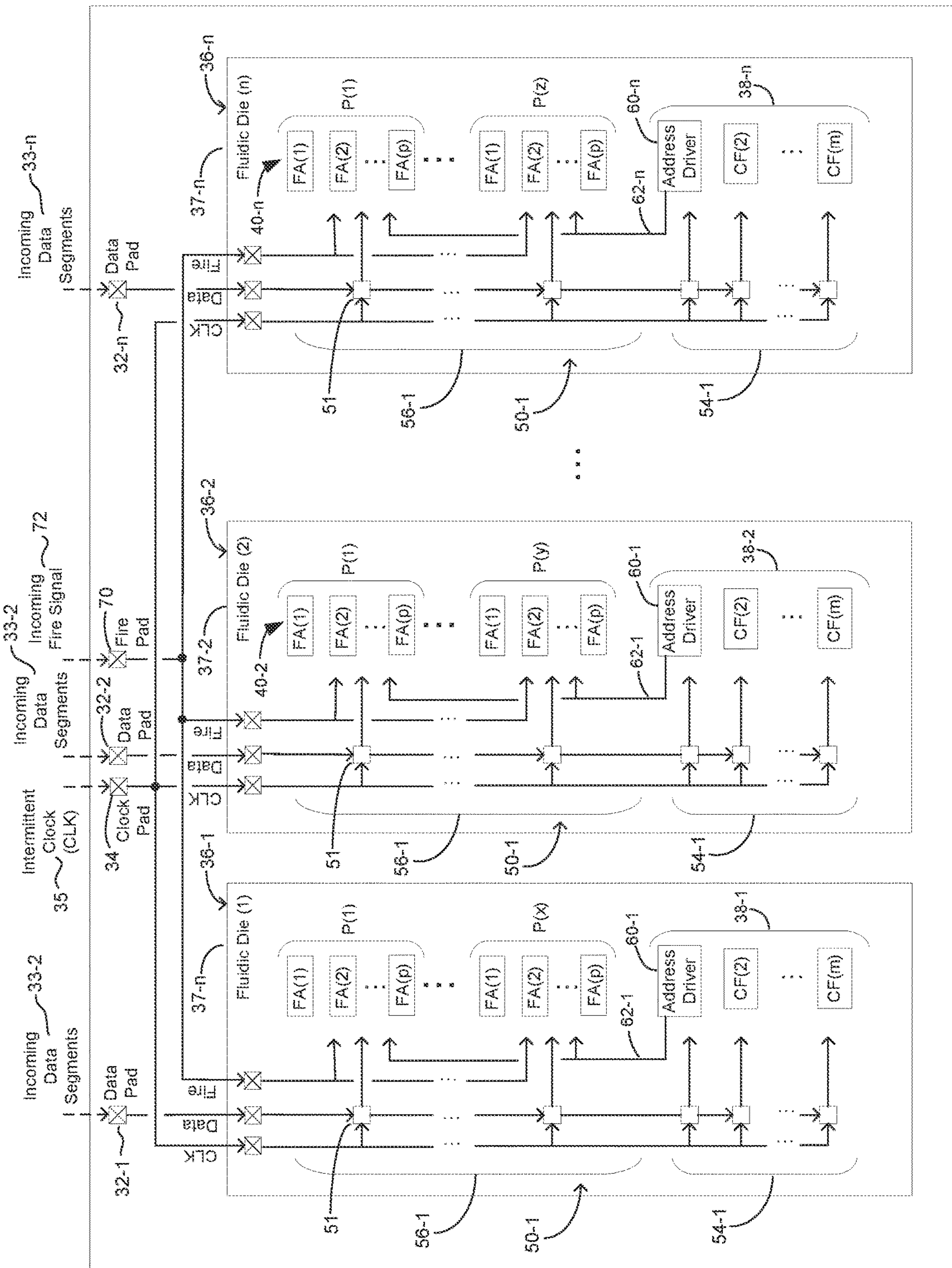
Hewlett-Packard Development Company, L.P., Int. Appl. No. PCT/US2019/016735 entitled Issue Determinations Responsive to Measurments filed Feb. 6, 2019 (29 pages).

* cited by examiner



Incoming Data Segments
33-n
Data Pad
32-n
52-n
CLK
58
36-n
Actuator Group (n)
40-n
FA(1)
FA(2)
FA(z)
38-n
CF(1)
CF(2)
CF(m)
50-n
51
56-n
51
54-n
30
Print Componet
Incoming Data Segments
33-2
32-2
Data Pad
52-2
Intermittent Clock (CLK)
35
34
Clock Pad
58
CLK
36-2
Actuator Group (2)
40-2
FA(1)
FA(2)
FA(y)
38-2
CF(1)
CF(2)
CF(m)
50-2
51
56-2
51
54-2
Incoming Data Segments
33-2
Data Pad
32-1
52-1
58
CLK
36-1
Actuator Group 1
40-1
FA(1)
FA(2)
FA(x)
38-1
CF(1)
CF(2)
CF(m)
50-1
51
56-1
51
54-1

Fig. 1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Data Pad
32-n
Incoming Data Segments
33-2
Data Pad
32-2
Incoming Fire Signal
72
Fire Pad
70
Intermittent Clock (CLK)
35
Clock Pad
34
Incoming Data Segments
33-2
Data Pad
32-1
Fluidic Die (n)
36-n
37-n
40-n
P(1)
P(z)
FA(1)
FA(2)
FA(p)
Address Driver
CF(2)
CF(m)
60-n
62-n
38-n
Fire
Data
CLK
51
56-1
50-1
54-1
Fluidic Die (2)
36-2
37-2
40-2
P(y)
60-1
62-1
38-2
Fluidic Die (1)
36-1
P(x)
38-1

Fig. 2

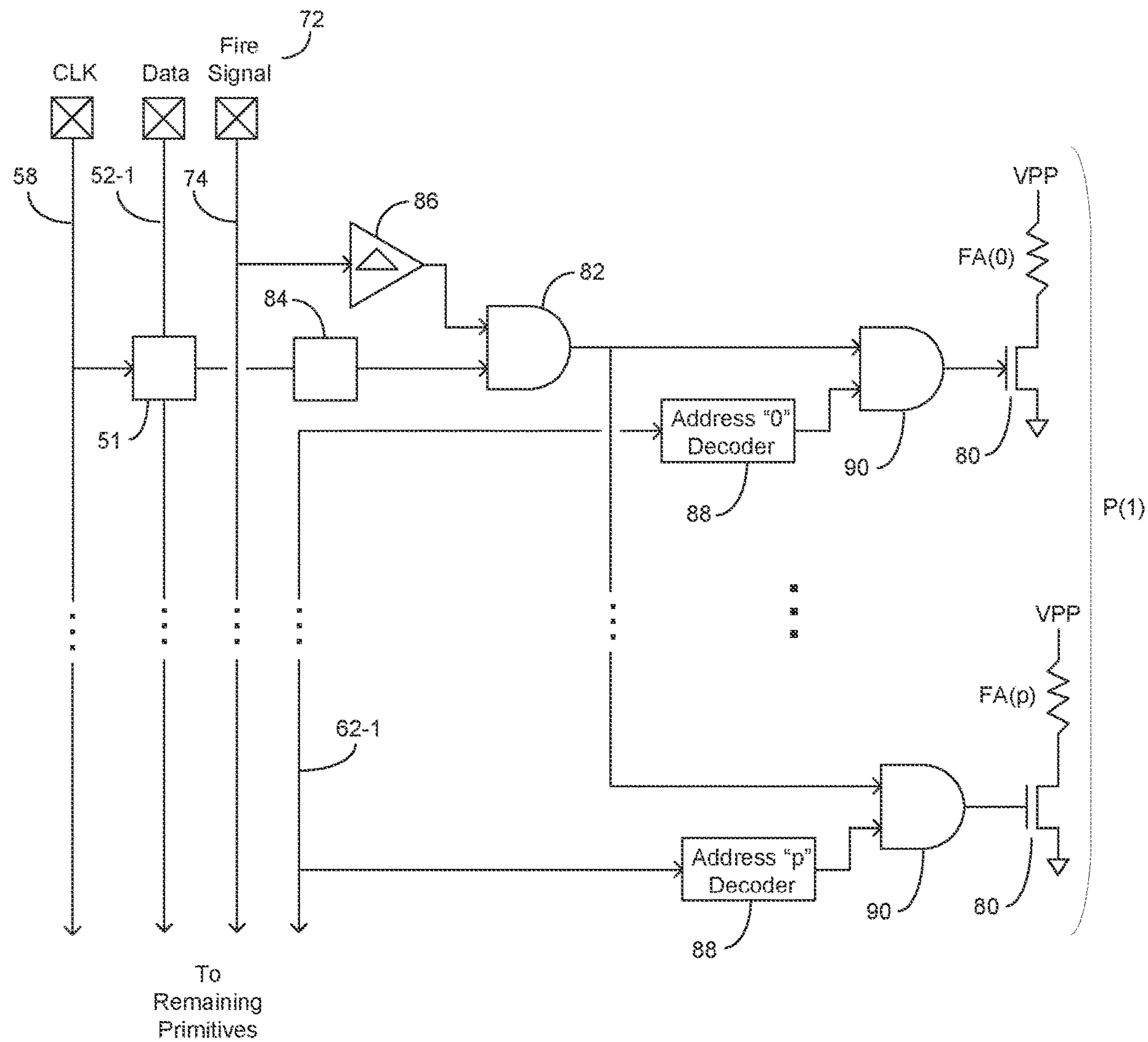
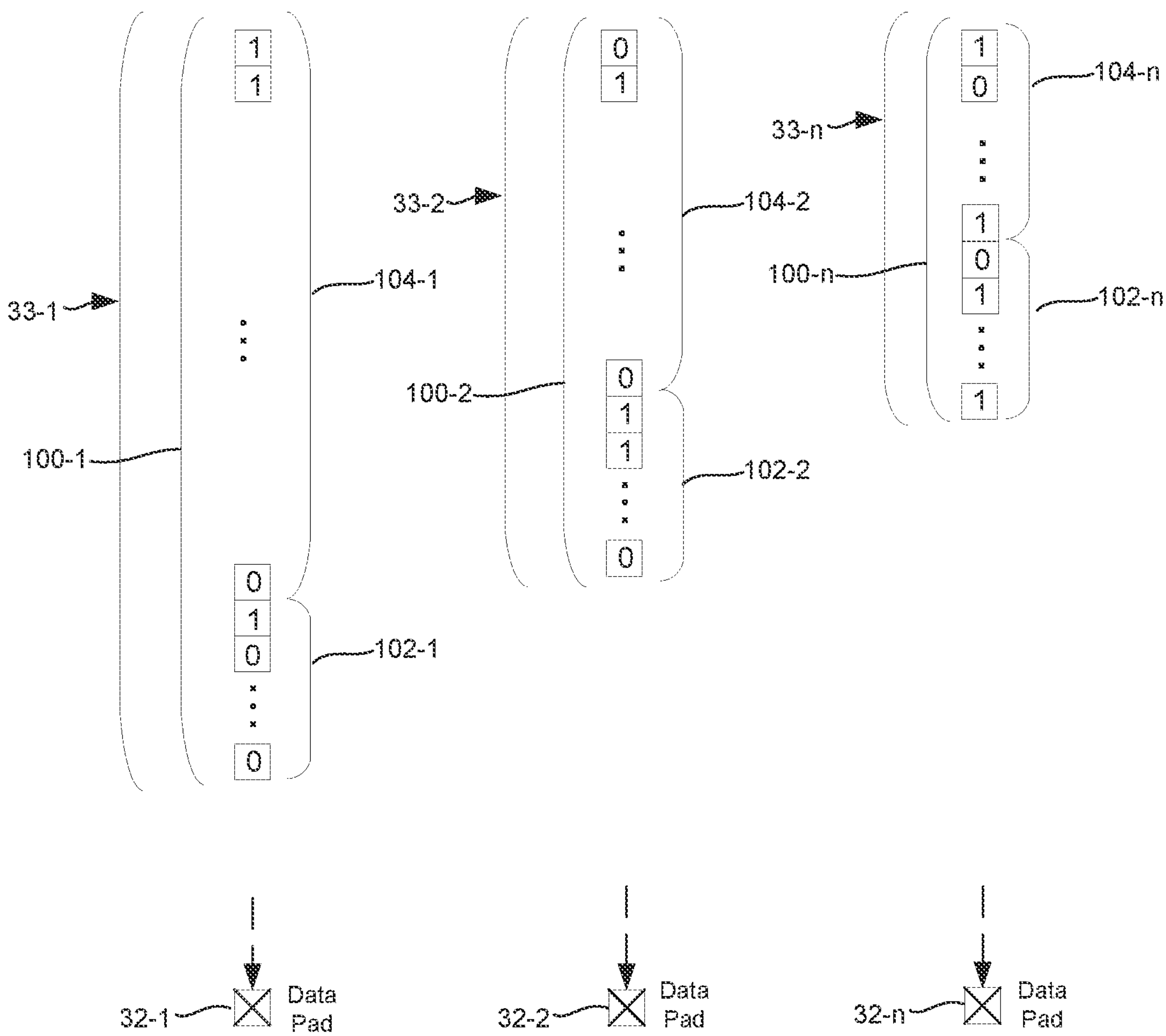
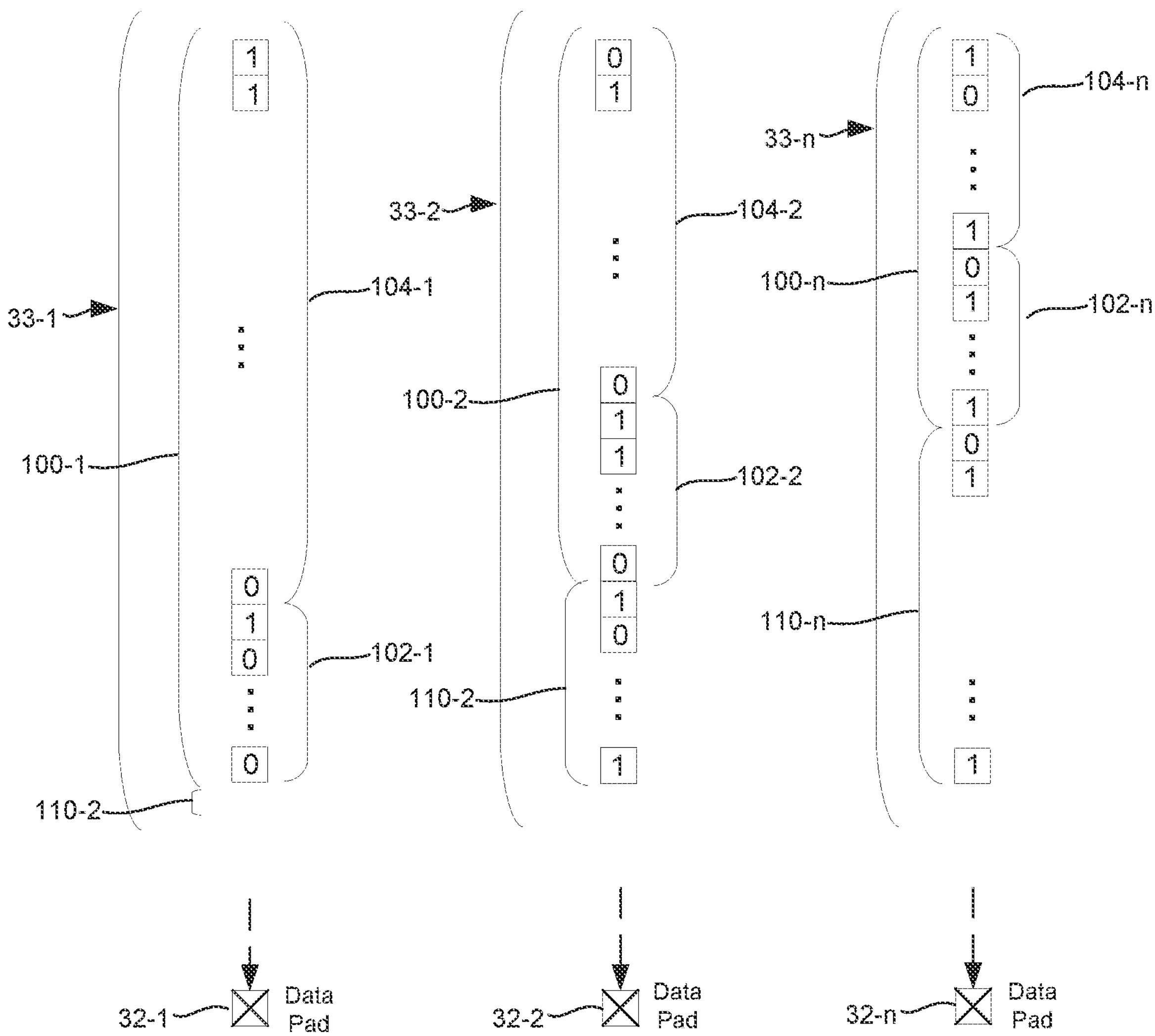

72
Fire Signal
CLK
Data
58
52-1
74
86
82
84
VPP
FA(0)
51
Address "0" Decoder
90
80
88
P(1)
VPP
FA(p)
62-1
Address "p" Decoder
90
80
88
To Remaining Primitives

Fig. 3



33-1
100-1
104-1
102-1
1
1
0
1
0
0
33-2
100-2
104-2
102-2
0
1
0
1
1
0
33-n
100-n
104-n
102-n
1
0
1
0
1
1
32-1
Data Pad
32-2
Data Pad
32-n
Data Pad

Fig. 4A



33-1
100-1
104-1
102-1
110-2
33-2
100-2
104-2
102-2
110-2
33-n
100-n
104-n
102-n
110-n
32-1
Data Pad
32-2
Data Pad
32-n
Data Pad

Fig. 4B

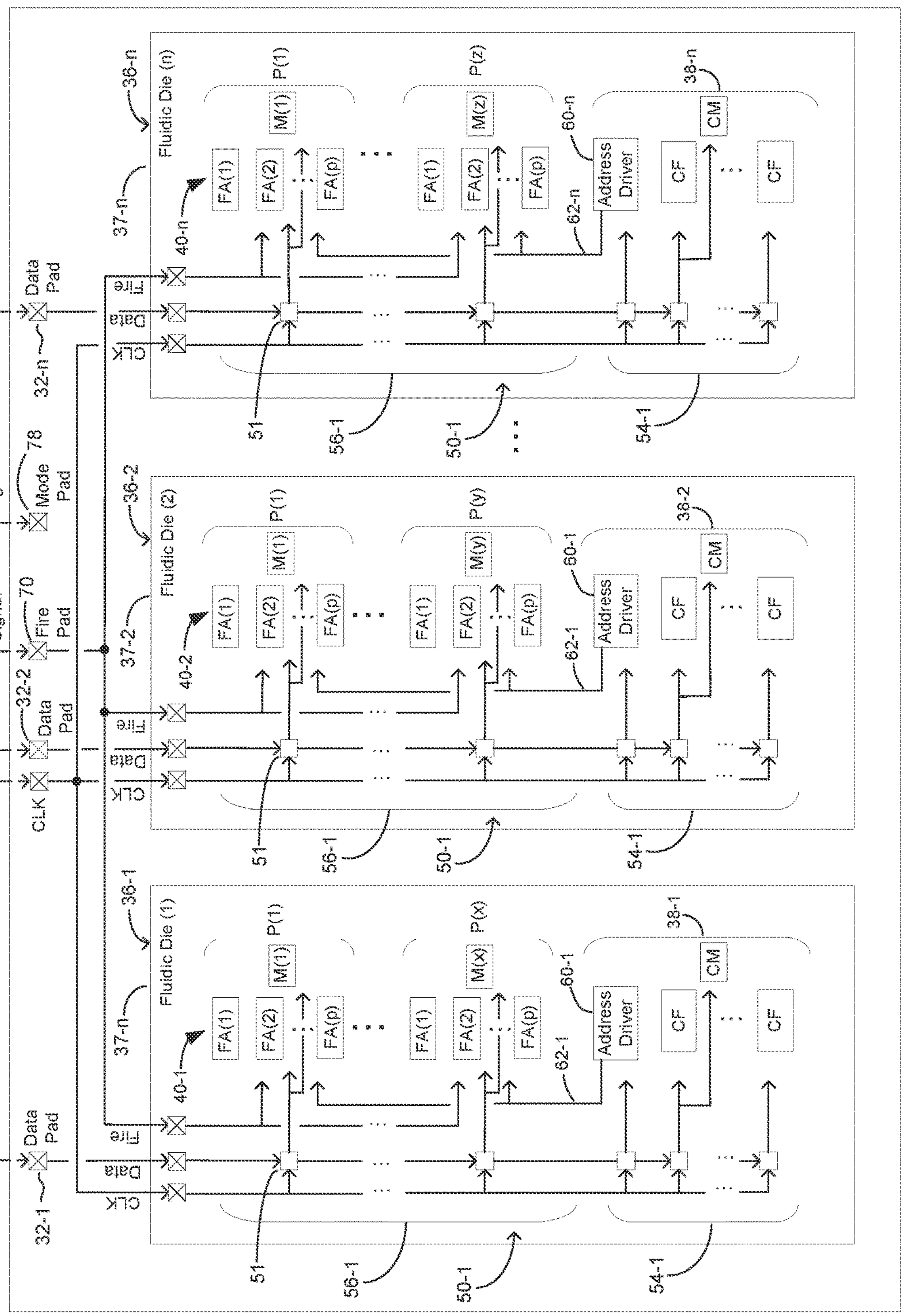

Incoming Data Segments
33-2
Data Pad
32-1
Intermittent Clock
35
CLK
Incoming Data Segments
33-2
Data Pad
32-2
Incoming Fire Signal
72
Fire Pad
70
Incoming Mode Signal
79
Mode Pad
78
Incoming Data Segments
33-n
Data Pad
32-n
CLK
Data
Fire
Fluidic Die (1)
36-1
37-n
40-1
FA(1)
FA(2)
FA(p)
M(1)
P(1)
M(x)
P(x)
Address Driver
60-1
62-1
CF
CM
38-1
51
56-1
50-1
54-1
Fluidic Die (2)
36-2
37-2
40-2
M(y)
P(y)
38-2
Fluidic Die (n)
36-n
37-n
40-n
M(z)
P(z)
60-n
62-n
38-n

Fig. 5

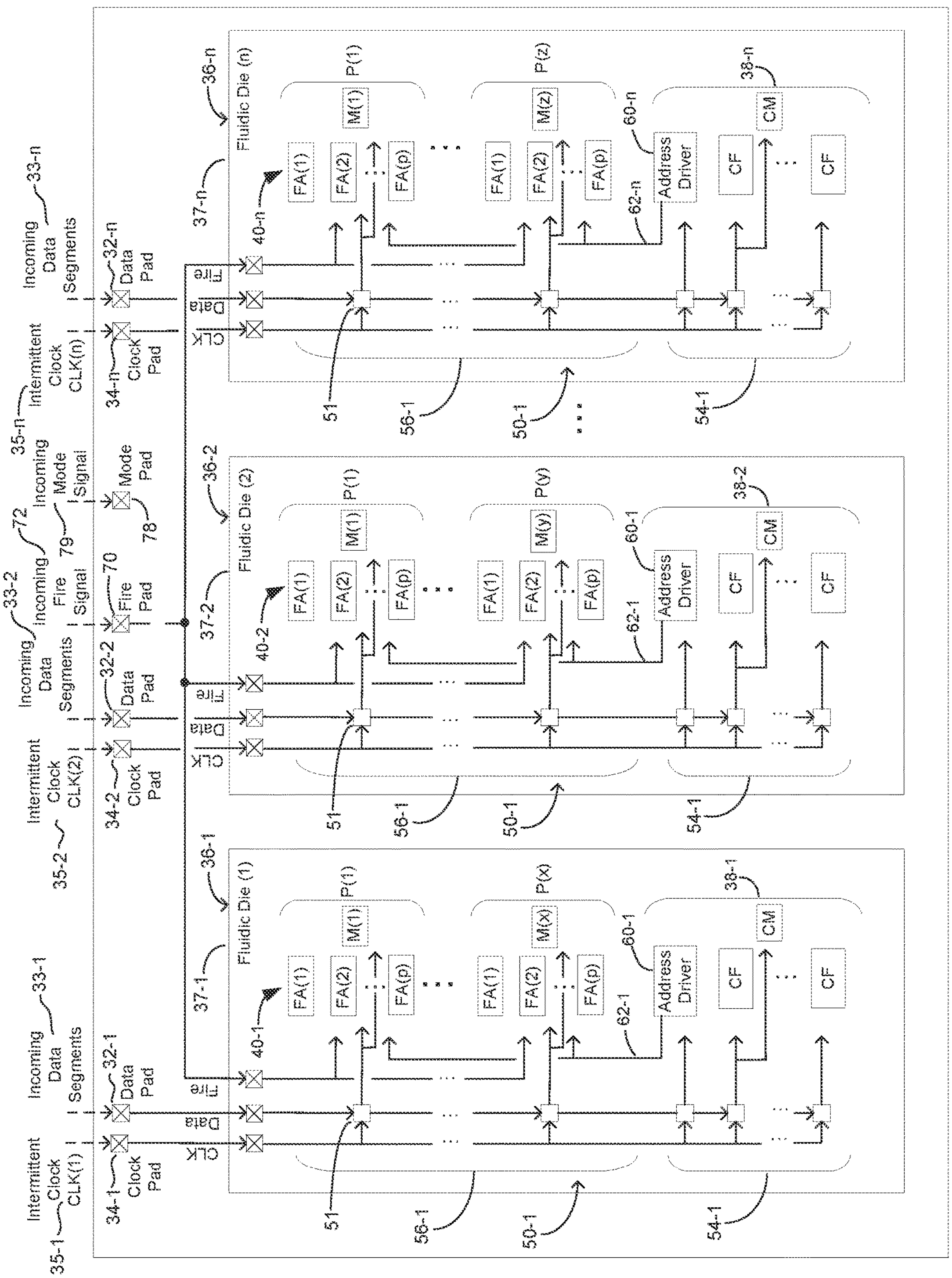

Intermittent Clock CLK(1)
35-1
34-1
Clock Pad
Incoming Data Segments
33-1
32-1
Data Pad
35-2
Intermittent Clock CLK(2)
34-2
Incoming Data Segments
32-2
33-2
Incoming Fire Signal
70
Fire Pad
72
Incoming Mode Signal
79
Mode Pad
78
35-n
Intermittent Clock CLK(n)
34-n
Incoming Data Segments
33-n
32-n
CLK
Data
Fire
Fluidic Die (1)
36-1
37-1
40-1
FA(1)
FA(2)
FA(p)
M(1)
P(1)
M(x)
P(x)
Address Driver
60-1
62-1
CF
CM
38-1
51
56-1
50-1
54-1
Fluidic Die (2)
36-2
37-2
40-2
M(y)
P(y)
38-2
Fluidic Die (n)
36-n
37-n
40-n
M(z)
P(z)
60-n
62-n
38-n

Fig. 6

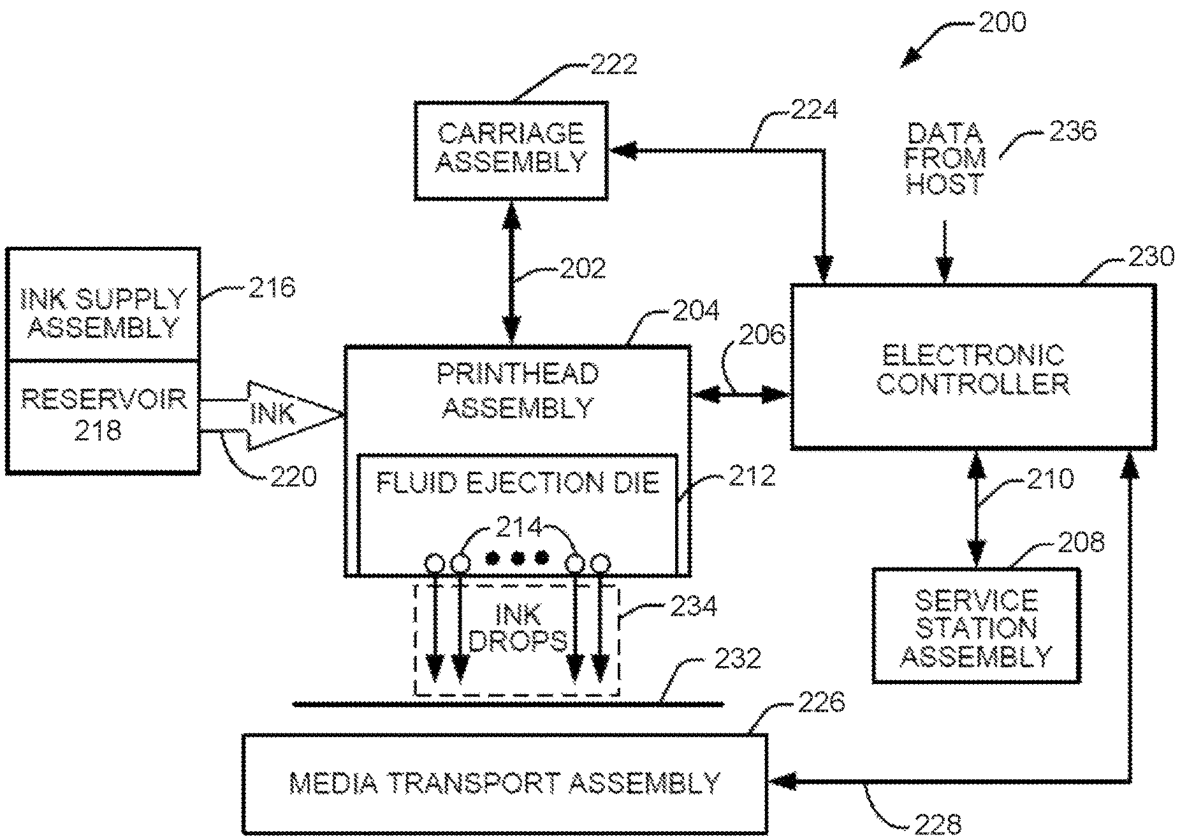

200
222
CARRIAGE ASSEMBLY
224
DATA FROM HOST
236
INK SUPPLY ASSEMBLY
216
202
230
204
206
ELECTRONIC CONTROLLER
PRINTHEAD ASSEMBLY
RESERVOIR 218
INK
220
FLUID EJECTION DIE
212
210
214
208
SERVICE STATION ASSEMBLY
INK DROPS
234
232
226
MEDIA TRANSPORT ASSEMBLY
228

Fig. 7

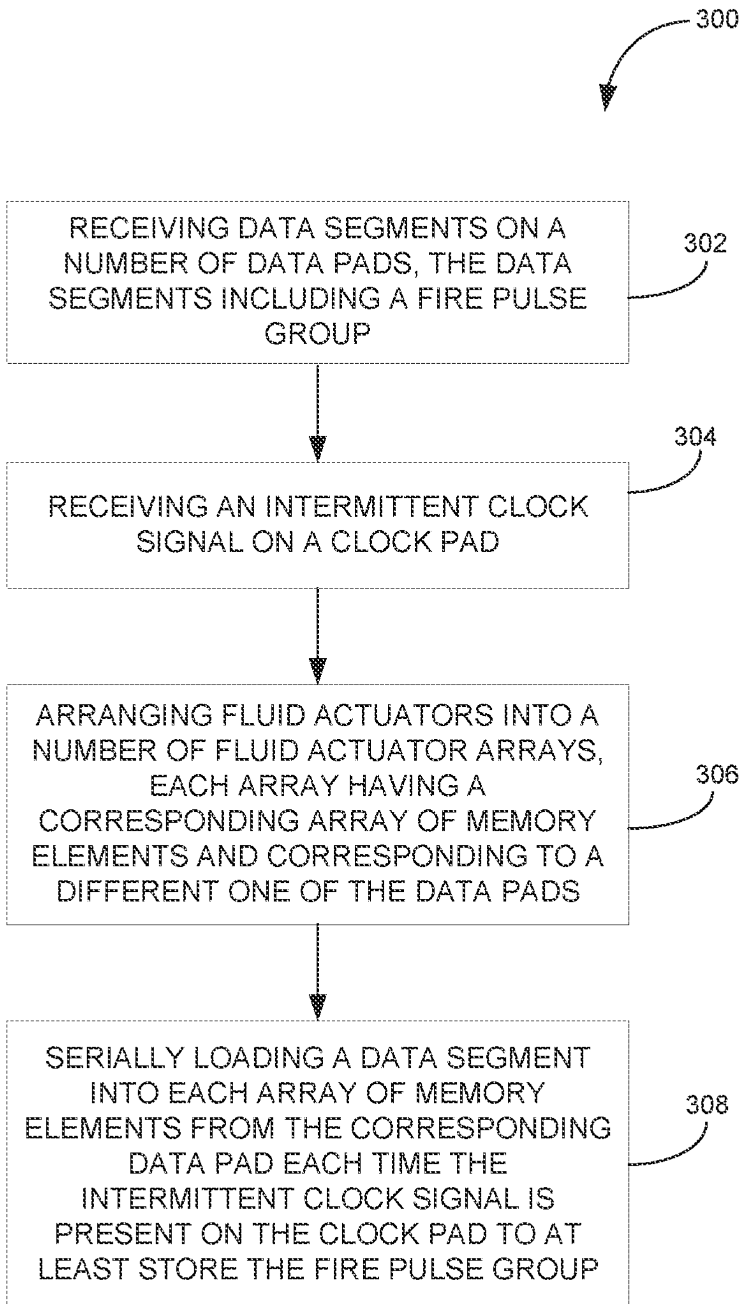

300
RECEIVING DATA SEGMENTS ON A NUMBER OF DATA PADS, THE DATA SEGMENTS INCLUDING A FIRE PULSE GROUP
302
RECEIVING AN INTERMITTENT CLOCK SIGNAL ON A CLOCK PAD
304
ARRANGING FLUID ACTUATORS INTO A NUMBER OF FLUID ACTUATOR ARRAYS, EACH ARRAY HAVING A CORRESPONDING ARRAY OF MEMORY ELEMENTS AND CORRESPONDING TO A DIFFERENT ONE OF THE DATA PADS
306
SERIALLY LOADING A DATA SEGMENT INTO EACH ARRAY OF MEMORY ELEMENTS FROM THE CORRESPONDING DATA PAD EACH TIME THE INTERMITTENT CLOCK SIGNAL IS PRESENT ON THE CLOCK PAD TO AT LEAST STORE THE FIRE PULSE GROUP
308

Fig. 8

PRINT COMPONENT WITH MEMORY ARRAY USING INTERMITTENT CLOCK SIGNAL

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a U.S. Continuation Application of National Stage application Ser. No. 16/767,914, filed May 28, 2020, entitled "PRINT COMPONENT WITH MEMORY ARRAY USING INTERMITTENT CLOCK SIGNAL" which is a U.S. National Stage Application of PCT Application No. PCT/US2019/016727, filed Feb. 6, 2019, entitled "PRINT COMPONENT WITH MEMORY ARRAY USING INTERMITTENT CLOCK SIGNAL".

BACKGROUND

Some print components may include an array of nozzles and/or pumps each including a fluid chamber and a fluid actuator, where the fluid actuator may be actuated to cause displacement of fluid within the chamber. Some example fluidic dies may be printheads, where the fluid may correspond to ink or print agents. Print components include printheads for 2D and 3D printing systems and/or other high precision fluid dispense systems.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a block and schematic diagram illustrating a print component, according to one example.

FIG. **2** is a block and schematic diagram illustrating a print component, according to one example.

FIG. **3** is a block and schematic diagram generally illustrating portions of a primitive arrangement, according to one example.

FIG. **4**A is a schematic diagram generally illustrating data segments, according to one example.

FIG. **4**B is a schematic diagram generally illustrating data segments, according to one example.

FIG. **5** is a block and schematic diagram illustrating a print component, according to one example.

FIG. **6** is a block and schematic diagram illustrating a print component, according to one example.

FIG. **7** is a schematic diagram illustrating a block diagram illustrating one example of a fluid ejection system.

FIG. **8** is a flow diagram illustrating a method of operating a print component, according to one example.

Throughout the drawings, identical reference numbers designate similar, but not necessarily identical, elements. The figures are not necessarily to scale, and the size of some parts may be exaggerated to more clearly illustrate the example shown. Moreover the drawings provide examples and/or implementations consistent with the description; however, the description is not limited to the examples and/or implementations provided in the drawings.

DETAILED DESCRIPTION

In the following detailed description, reference is made to the accompanying drawings which form a part hereof, and in which is shown by way of illustration specific examples in which the disclosure may be practiced. It is to be understood that other examples may be utilized and structural or logical changes may be made without departing from the scope of the present disclosure. The following detailed description, therefore, is not to be taken in a limiting sense, and the scope of the present disclosure is defined by the appended claims. It is to be understood that features of the various examples described herein may be combined, in part or whole, with each other, unless specifically noted otherwise.

Examples of fluidic dies may include fluid actuators. The fluid actuators may include thermal resistor based actuators (e.g. for firing or recirculating fluid), piezoelectric membrane based actuators, electrostatic membrane actuators, mechanical/impact driven membrane actuators, magneto-strictive drive actuators, or other suitable devices that may cause displacement of fluid in response to electrical actuation. Fluidic dies described herein may include a plurality of fluid actuators, which may be referred to as an array of fluid actuators. An actuation event may refer to singular or concurrent actuation of fluid actuators of the fluidic die to cause fluid displacement. An example of an actuation event is a fluid firing event whereby fluid is jetted through a nozzle.

In example fluidic dies, the array of fluid actuators may be arranged in sets of fluid actuators, where each such set of fluid actuators may be referred to as a "primitive" or a "firing primitive." The number of fluid actuators in a primitive may be referred to as a size of the primitive. In some examples, the set of fluid actuators of each primitive are addressable using a same set of actuation addresses, with each fluid actuator of a primitive corresponding to a different actuation address of the set of actuation addresses, with the addresses being communicated via an address bus. In some examples, a fluidic actuator of a primitive will actuate (e.g., fire) in response to a fire signal (also referred to as a fire pulse) based on actuation data corresponding to the primitive (sometimes also referred to as nozzle data or primitive data) when the actuation address corresponding to the fluidic actuator is present on the address bus.

In some cases, electrical and fluidic operating constraints of a fluidic die may limit which fluid actuators of each primitive may be actuated concurrently for a given actuation event. Primitives facilitate addressing and subsequent actuation of fluid actuator subsets that may be concurrently actuated for a given actuation event to conform to such operating constraints.

To illustrate by way of example, if a fluidic die comprises four primitives, with each primitive including eight fluid actuators (with each fluid actuator corresponding to a different address of a set of addresses 0 to 7), and where electrical and fluidic constraints limit actuation to one fluid actuator per primitive, a total of four fluid actuators (one from each primitive) may be concurrently actuated for a given actuation event. For example, for a first actuation event, the respective fluid actuator of each primitive corresponding to address "0" may be actuated. For a second actuation event, the respective fluid actuator of each primitive corresponding to address "5" may be actuated. As will be appreciated, such example is provided merely for illustration purposes, with fluidic dies contemplated herein may comprise more or fewer fluid actuators per primitive and more or fewer primitives per die.

Example fluidic dies may include fluid chambers, orifices, and/or other features which may be defined by surfaces fabricated in a substrate of the fluidic die by etching, microfabrication (e.g., photolithography), micromachining processes, or other suitable processes or combinations thereof. Some example substrates may include silicon based substrates, glass based substrates, gallium arsenide based substrates, and/or other such suitable types of substrates for microfabricated devices and structures. As used herein, fluid chambers may include ejection chambers in fluidic communication with nozzle orifices from which fluid may be ejected, and fluidic channels through which fluid may be conveyed. In some examples, fluidic channels may be microfluidic channels where, as used herein, a microfluidic channel may correspond to a channel of sufficiently small size (e.g., of nanometer sized scale, micrometer sized scale, millimeter sized scale, etc.) to facilitate conveyance of small volumes of fluid (e.g., picoliter scale, nanoliter scale, microliter scale, milliliter scale, etc.).

In some examples, a fluid actuator may be arranged as part of a nozzle where, in addition to the fluid actuator, the nozzle includes an ejection chamber in fluidic communication with a nozzle orifice. The fluid actuator is positioned relative to the fluid chamber such that actuation of the fluid actuator causes displacement of fluid within the fluid chamber that may cause ejection of a fluid drop from the fluid chamber via the nozzle orifice. Accordingly, a fluid actuator arranged as part of a nozzle may sometimes be referred to as a fluid ejector or an ejecting actuator.

In some examples, a fluid actuator may be arranged as part of a pump where, in addition to the fluidic actuator, the pump includes a fluidic channel. The fluidic actuator is positioned relative to a fluidic channel such that actuation of the fluid actuator generates fluid displacement in the fluid channel (e.g., a microfluidic channel) to convey fluid within the fluidic die, such as between a fluid supply and a nozzle, for instance. An example of fluid displacement/pumping within the die is sometimes also referred to as microrecirculation. A fluid actuator arranged to convey fluid within a fluidic channel may sometimes be referred to as a non-ejecting or microrecirculation actuator. In one example nozzle, the fluid actuator may comprise a thermal actuator, where actuation of the fluid actuator (sometimes referred to as "firing") heats the fluid to form a gaseous drive bubble within the fluid chamber that may cause a fluid drop to be ejected from the nozzle orifice. As described above, fluid actuators may be arranged in arrays (such as columns), where the actuators may be implemented as fluid ejectors and/or pumps, with selective operation of fluid ejectors causing fluid drop ejection and selective operation of pumps causing fluid displacement within the fluidic die. In some examples, the array of fluid actuators may be arranged into primitives.

Some printheads receive data in the form of data packets, sometimes referred to as fire pulse groups or a fire pulse group data packets, where each data packet includes a head portion and a body portion. In some examples, the head portion includes a sequence of start bits and configuration data for on-die functions such as address bits for address drivers, and fire pulse data for fire pulse selection, for example. The body portion of the packet includes primitive data, such as actuator data and/or memory data, that selects which nozzles corresponding to address represented by the address bits in the primitives will be actuated (or fired) and, in some examples, represents data to be written to memory elements of memory arrays associated the primitives. The fire pulse group data pack concludes with stop bits indicating the end of the data packet.

Such printheads include data parsers which use a free-running clock and operate to capture incoming data bits as they are received by the printhead in order to detect the start pattern and thereby identify the beginning of a fire pulse group data packet. Upon detecting a start pattern, the data parser circuitry collects bits as they are received and directs them to the appropriate primitives. In some examples, to determine when the data packet is complete, the data parser circuitry counts the total number of bits received. When the correct number of bits for a data packet has been received, the data parser circuitry stops distributing bits and returns to monitoring incoming data to identify a start sequence for another data packet.

Among other functions, data parser circuitry typically includes several counters, such as to indicate a particular group of primitives to which the data is to be directed (e.g., a printhead may include multiple columns of primitives), and to count a total number bits which have been received, for example. Data parser circuitry consumes relatively large amounts of silicon area on a printhead die, thereby increasing the size and cost of the die. Additionally, data parser circuitry is inflexible and requires each fire pulse group data packet for a printhead to have a fixed length. Additionally, a free running clock can potentially introduce electromagnetic interference (EMI) issues to the die.

The present disclosure, as will be described in greater detail herein, provides a print component having an array of memory elements to serially receive a segment of data bits including configuration data and primitive data each time an intermittent clock signal is received on a clock pad, which eliminates data parser circuitry and a free running clock. Such an arrangement reduces silicon area requirements, eliminates EMI introduced by a free-running clock signal, and enables arrays of fluid actuators having different primitive sizes, such as different fluidic dies, to share a clock and fire signals, which reduces interconnect complexity.

FIG. **1** is a block and schematic diagram generally illustrating a print component **30**, according to one example of the present disclosure, including a plurality of data pads **32**, illustrated as data pads **32-1** to **32**-N, a clock pad **34** to receive an intermittent clock signal **35**, and a plurality of actuator groups **36**, illustrated as actuator groups **36-1** to **36**-N, with each actuator group **36** corresponding to a different one of the data pads **32**. In one example, each of the actuator groups **36** corresponds to a different fluid type. For instance, in one case, print component **30** comprises a printhead with each actuator group corresponding to a different type of ink (e.g., black, cyan, magenta, and yellow). In one example, each actuator group **36** of print component **30** is implemented in a different respective fluidic die where, in one case, each respective fluidic die corresponds to a different liquid type.

According to one example, each actuator group **36** includes a group of configuration functions **38**, illustrated as **38-1** to **38**-N, an array of fluid actuators **40**, illustrated as arrays **40-1** to **40**-N, and an array of memory elements **50**, illustrated as arrays **50-1** to **50**-N. In one case, each group of configuration functions **38** includes a number of configuration functions, illustrated as configuration functions CF(**1**) to CF(m), for configuring an operational setup of the corresponding actuator group **36**. In examples, configuration functions CF(**1**) to CF(m) may include functions such as an address driver, a fire pulse configuration function, and a sensor configuration function (e.g., thermal sensors), for instance.

In one example, each array of fluid actuators **40** includes a number of fluid actuators (FAs), with array **40-1** of actuator group **36-1** including fluid actuators FA(**1**) to FA(x), array **40-2** of actuator group **36-2** including fluid actuators FA(**1**) to FA(y), and array **40**-N of actuator group **40**-N including fluid actuators FA(**1**) to FA(z). In one case, each array of fluid actuators **40** may have a same number of fluid actuators (x=y=z). In other cases, the arrays of fluid actuators **40** may have differing numbers of fluid actuators (x≠y≠z).

The array of memory elements **50** of each actuator group **36** comprises a number of memory elements **51**, with each array **50** having a first portion of memory elements **52**, illustrated as first portions **52-1** to **52**-N, corresponding to the respective group of configuration functions **38**, and a second portion of portion of memory elements **54**, illustrated as second portions **56-1** to **56**-N, corresponding to the respective array of fluid actuators **40**. In some cases, the array of memory elements **50** of each actuator group **36** may have a same number of memory elements **51**. In other cases, the array of memory elements **50** of different actuator groups **36** may have different numbers of memory elements **51**.

The array of memory elements **50** of each actuator group **36** is connected to the corresponding data pad **32** via a corresponding communication path **52**, with the arrays of memory elements **50-1** to **50**-N being respectively connected to data pads **32-1** to **32**-N by communication paths **52-1** to **52**-*n*. In one example, as illustrated by the arrangement of FIG. **1**, each array of memory elements **50** of each group of fluid actuators **36** is connected to and receives intermittent clock signal **35** via clock pad **34**.

In one example, each time intermittent clock **35** is present on clock pad **34** of print component **30**, the array of memory elements **50** of each actuator group **36** serially loads a data segment **33** comprising a series of data bits from the corresponding data pad **32**, illustrated as data segments **33-1** to **33**-*n*, with the data bits loaded into the first portion of memory elements **52** and into the second portion of memory elements **54** respectively corresponding to the group of configuration functions **38** and to the array of fluid actuators **40**. In one example, each time intermittent clock signal **35** is present on clock pad **34**, the array of memory elements **50** of each actuator group **36** serially loads the series of data bits of a current data segment **33**, which replace the previously loaded data bits of the preceding data segment **33**.

In one example, as will be described in greater detail below (e.g., see FIG. **3**), the series of data bits of each data segment **33** include fire pulse groups similar to that described above. However, because print component **30**, loads each data segment **33** only when intermittent clock signal **35** is present on clock pad **34** (i.e., does not employ a free running clock), the fire pulse groups of data segments **33** do not include a start-bit sequence. Since data segments **33** do not include a start-bit sequence and are loaded into the array of memory elements **50** only when intermittent clock signal **35** is present on clock pad **34**, print component **30** and actuator groups **36**, in accordance with the present disclosure, do not include data parser circuitry, thereby saving circuit area and reducing costs.

Additionally, as described in greater detail below, using an intermittent clock signal **35** and an array of memory elements **50** to serially receive data enables print component **30** to support multiple arrays of fluid actuators **40** having differing numbers of fluid actuators and using fire pulse groups of varying lengths while operating on a same intermittent clock signal **35** and sharing a common fire signal (as will be described in greater detail below). Furthermore, employing an intermittent clock signal eliminates potential EMI problems associated with free-running clocks.

FIG. **2** is a block and schematic diagram generally illustrating a print component **30**, according to one example of the present disclosure. In one example, the actuator groups **36-1** to **36**-*n* are implemented as fluidic dies **37-1** to **37**-*n*. According to the example of FIG. **2**, the fluid actuators (FA) of each of the arrays of fluid actuators **40-1** to **40**-*n* of actuator groups **36-1** to **36**-*n* are arranged to form a number of primitives, with the fluid actuators of array **40-1** of actuator group **36-1** arranged to form primitive P(**1**) to P(x), the fluid actuators of array **40-2** of actuator group **36-2** arranged to form primitive P(**1**) to P(y), and the fluid actuators of array **40**-*n* of actuator group **36**-*n* arranged to form primitive P(**1**) to P(z), with each primitive including a number of fluid actuators FA(**1**) to FA(p). In one case, each array of fluid actuators **40** may have a same number of primitives (x=y=z). In other cases, the arrays of fluid actuators **40** may have differing numbers of primitives (x≠y≠z). Although the primitives of each actuator group **36** is illustrated as having a same number of fluid actuators, p, in other examples, the number of fluid actuators in each primitive may vary between actuator groups **36**.

In one example, as illustrated, the array of memory elements **50** of each actuator group **37** comprises a series or chain of memory elements **51** implemented to function as a serial-to-parallel data converter, with first portion **54** of memory elements **51** corresponding to the group of configuration functions **38**, and second portion of memory elements **56** corresponding to the array of fluid actuators **40**, with each memory element **51** o the second portion **56** corresponding to a different one of the primitives P(**1**) to P(x). In one example, the array of memory elements **50** of each actuator group **36** comprises a sequential logic circuit (e.g., flip-flop arrays, latch arrays, etc.). In one example, the sequential logic circuit is adapted to function as a serial-in, parallel-out shift register.

According to one example, the group of configuration functions **38** of each actuator group **36** includes an address driver **60**, illustrated at address drivers **60-1** to **60**-*n*, which drives an address onto a corresponding address bus **62**, illustrated as address buses **62-1** to **62**-*n*, based on address bits in corresponding memory elements **51** of first portion **54** of the array of memory elements **50**, with memory bus **62** communicating the driven address to fluid actuators FA(**1**) to FA(p) of each of the corresponding primitives. In one example, print component **30** includes a fire pad **70** to receive a fire signal **72** which is communicated to each of the actuator groups **36** via a communication path **74**.

An example of the operation of print component **30** of FIG. **2** is described below with reference to FIGS. **3** and **4**. FIG. **3** is a block and schematic diagram generally illustrating portions of a primitive arrangement for the primitives of actuator groups **36-1** to **36**-*n* of FIG. **2**. For illustrative purposes, the block and schematic diagram of FIG. **2** is described with reference to primitive P(**1**) of actuator group **36-1** of FIG. **2**.

In example, each fluid actuator, illustrated as a thermal resistor in FIG. **3**, is connectable between a power source, VPP, and a reference potential (e.g., ground) via a corresponding controllable switch, such as illustrated by FETs **80**.

According to one example, each primitive, including primitive P(**1**), includes an AND-gate **82** receiving, at a first input, primitive data (e.g., actuator data) for primitive P(**1**) stored in a local memory element **84**, where local memory element receives such primitive data from corresponding memory element **51** of the array of memory elements **50-1** of actuator group **36-1**. At a second input, AND-gate **82** receives fire signal **72** via communication path **70**. In one example, fire signal **72** is delayed by a delay element **86**, with each primitive having a different delay so that firing of fluid actuators is not simultaneous among primitives P(**1**) to P(x).

In one example, each fluid actuator has a corresponding address decoder **88** receiving the address driven by address driver **60-1** on address bus **62-1**, and an AND-gate **90** for controlling a gate of FET **80**. AND-gate **90** receives the output of corresponding address decoder **88** at a first input, and the output of AND-gate **82** at a second input. It is noted that address decoder **88** and AND-gate **90** are repeated for each fluid actuator, while AND-gate **82**, memory element **84**, and delay element **86** are repeated for each primitive.

FIG. **4**A is a block diagram generally illustrating example data segments **33-1** to **33**-*n* respectively received by print component **30** via data pads **32-1** to **32**-*n*. As illustrated, each data segment **33** includes a fire pulse group **100** including a first portion of data bits **102** corresponding to the group of configuration functions **38** (sometimes referred to as configuration data), and a second portion of data bits **104** corresponding to the array of fluid actuators **40** (sometimes referred to as primitive data). For instance, with respect to data segment **33-1**, the data bits of the first portion of data bits **102-1** correspond to the group of configuration functions **38-1** and include address data bits for address driver **60-1**, and the data bits of the second portion of data bits **104-1** correspond to the array of fluid actuators **40-1**, with each data bit of second portion **104-1** corresponding to a different one of the primitives P(**1**) to P(x). For each data segment **33**, the number of data bits of the fire pulse group **32** (i.e., the number of fire pulse bits) is equal to the sum of the number of bits of the first portion of data bits **102** (i.e., configuration data bits) and the number of bits of the second portion of data bits **104** (i.e., primitive data).

According to the example of FIG. **4**A, second portion **104-1** of fire pulse group **100-1** of data segment **33-1** is illustrated as having more primitive data bits than second portion **104-2** of fire pulse group **100-2** of data segment **33-2**, and second portion **104-2** of fire pulse group **100-2** of data segment **33-2** is illustrated as having more primitive data bits than second portion **104**-*n* of fire pulse group **100**-*n* of data segment **33**-*n*, meaning that, with reference to FIG. **2**, the array of fluidic actuators **40-1** of fluidic die **36-1** has a greater number of primitives than the array of fluidic actuators **40-2** of fluidic die **36-2**, while the array of fluidic actuators **40-2** of fluidic die **36-2** has a greater number of primitives than the array of fluidic actuators **40**-*n* of fluidic die **36**-*n* (i.e., x>y>z). As a result, fire pulse group **100-1** has more fire pulse group bits than fire pulse group **100-2**, and fire pulse group **100-2** has more fire pulse group bits than fire pulse group **100**-*n*, meaning that data segment **33-1** is longer (i.e., has more data segment bits) than data segment **33-2**, and that data segment **33-2** is longer (i.e., has more data segment bits) than data segment **33**-*n*.

With reference to FIG. **2**, upon intermittent clock signal **35** being received at clock pad **34** (e.g., upon receiving the first rising edge of intermittent clock signal **35**), data segments **33-1** to **33**-*n* are serially loaded into the memory elements **51** of their respective arrays of memory elements **50-1** to **50**-*n* of actuator groups **36-1** to **36**-*n*. However, when sharing a same intermittent clock signal **35**, as illustrated by the example implementation of FIG. **2**, because of their different lengths, the number of cycles of intermittent clock signal **35** needed to load fire pulse group **100-1** of data segment **33-1** into array of memory elements **50-1** is greater than a number of clock cycles needed to load fire pulse groups **100-2** and **100**-*n* of data segments **33-2** and **33**-*n* into their respective arrays of memory elements **50-2** and **50**-*n*. As a result, data bits of fire pulse groups **100-2** and **100**-*n* of data segments **33-2** and **33**-*n* will begin being respectively shifted out of arrays of memory elements **50-2** and **50**-*n* before data bits of fire pulse group **100-1** of data segment **33-1** have finished being serially loaded into the array of memory elements **50-1**. Consequently, if not accounted for, incorrect data will be populating the memory elements of arrays **50-2** and **50**-*n* upon completion of loading data segment **33-1** into array **50-1**.

With reference to FIG. **4**B, according to one example, when sharing an intermittent clock signal, such as clock signal **35**, in order to make each of the data segments **33-1** to **33**-*n* equal in length (i.e., a same number of bits) so as to take a same number of clock cycles of intermittent clock signal **35** to load into their respective memory arrays **50-1** to **50**-*n*, in addition to fire pulse groups **100-2** and **100**-*n*, data segments **33-1** and **33**-*n* each include a pre-pended segment of filler bits **110-1** and **110**-*n*. According to one example, as illustrated, since data segment **33-1** is the longest data segment (i.e., has the most segment bits), the segment of filler bits **110-1** of data segment **33-1** contains no filler bits, while segments of filler bits **110-2** and **110**-*n* each have a number of filler bits to respectively make data segments **33-2** and **33**-*n* the same length as data segment **33-1** (with filler bit segment **33**-*n* having more filler bits than filler bit segment **33-2**). According to the example illustration of FIG. **4**B, in general, segments of filler bits **110** are added to each shorter data segment **33** of data segments **33-1** to **33**-*n* so that all data segments **33-1** to **33**-*n* have a length the same as the longest data segment **33** of data segments **33-1** to **33**-*n*.

By pre-pending filler bit segments **110-1** to **110**-*n* to data segments **33-1** and **33**-*n*, in a case where an intermittent clock signal is shared by actuator groups **36-1** to **36**-*n*, when serially loading data segments **33-1** to **33**-*n* into their respective arrays of memory elements **50-1** to **50**-*n*, the last data bit of each data segments **33-1** to **33**-*n* will be loaded on the same clock cycle so that each fire pulse group is properly loaded into their respective memory array **50-1** to **50**-*n*, with the first and second portions of data bits **102** and **104** being respectively loaded into first and second portions **54** and **56** of the corresponding array of memory elements **50**.

Prepending filler bit segments **110** to at least data segments **33** having shorter lengths so that all data segments **33** have a same length enables a clock signal **35** to be shared by multiple arrays of fluidic actuators **36** even when such arrays of fluidic actuators **36** have differing numbers of fluid actuators (FAs), which reduces and simplifies circuitry, such as that of print component **30**.

In some examples, each of the data segments **33-1** to **33**-*n* includes a filler bit segment **100** including a number of filler bits, where the number of filler bits in each filler bit segment **100-1** to **100**-*n* is such that each of the data segments **33-1** to **33**-*n* has a same length. In one example, each of the filler bits has either a logic "high" value (e.g. "1") or a logic "low" value ("0"), where the filler bits of each filler bit segment **100** have a pattern of logic "low" and logic "high" values to mitigate electromagnetic effects on print component **30** as data segments **33-1** to **33**-*n* are respectively serially loaded in memory arrays **50-1** to **50**-*n*.

Continuing with the illustrative example above, referring to FIGS. **2-3**, in one case, upon the final data bit of each of the data segments **33-1** to **33**-*n* being loaded into the respective array of memory elements **50-1** to **50**-*n* (e.g., the last data bit each of the second portions **104-1** to **104**-*n* of fire pulse groups **100-1** to **100**-*n* being loaded into their respective memory element **51** corresponding to primitive P(**1**)), intermittent clock signal **35** is removed from clock pad **34** so that serial loading of data into memory arrays **50-1** to **50**-*n* ceases.

According to one example, upon completion of loading of fire pulse groups **100-1** to **100**-*n* into their respective memory arrays **50-1** to **50**-*n*, a fire signal **72** (e.g., a fire pulse signal) is received on fire pad **70**. With reference to FIGS. **2** and 3, in one example, in response to receipt of fire pulse signal 72, data stored in each memory element 51 of each array of memory elements 50-1 to 50-*n* are parallel shifted into a corresponding memory element in the corresponding array of fluid actuators 40-1 to 40-*n* or the group configuration functions 38-1 to 38-*n*. For example, in FIG. 3, in response to fire signal 72, primitive data stored in memory element 51 is shifted to a corresponding memory element 84 in primitive P(1).

In one example, after being parallel shifted out of the arrays of memory elements 50-1 to 50-*n*, the fire pulse group data is processed by the corresponding groups of configuration functions 38-1 to 38-*n* and primitives (P(1) to P(x), P(1) to P(y), and P(1) to P(z)) to operate selected fluid actuators (FAs) to circulate fluid or eject fluid drops. For instance, with reference to FIG. 3, in one example, if the primitive data stored in memory element 84 has a logic high (e.g., "1") and a fire pulse signal 72 is present on communication path 74, the output of AND-gate 82 is set to a logic "high". If the address driven on address bus 62-1 by address encoder 60-1 in response to the address bits received from the corresponding memory element of the second group of memory elements 54-1 represents address "0", the output of Address Decoder "0" 88 is set to a logic "high". With the output of AND-gate 82 and Address Decoder "0" 88 each set to a logic "high", the output of AND-gate 90 is also set to a logic "high", thereby turning "on" corresponding FET 80 to energize fluid actuator FA(0) to displace fluid (e.g., eject a fluid drop).

In one example, upon the fire pulse group data being shifted out of the arrays of memory elements 50-1 to 50-*n* in response to fire signal 72, intermittent clock signal 35 is again received via clock pad 34 and next data segments 33-1 to 33-*n* are serially loaded into the arrays of memory elements 50-1 to 50-*n*.

FIG. 5 is a block and schematic diagram generally illustrating print component 30 of FIG. 2, where in addition to fluid actuators FA(1) to FA(p), primitives P(1) to P(x), P(1) to P(y), and P(1) to P(z) of actuator groups 40-1 to 40-*n* each include an array of memory elements, respectively illustrated as M(1) to M(x), M(1) to M(y), and M(1) to M(z). In one example, as illustrated, each of the groups of configurations 38-1 to 38-*n* may include one or more memories, CM, each corresponding to a different one of the configuration functions.

In one example, print component 30 of FIG. 5 further includes a mode pad 78 to receive a mode signal 79. In one example, based on a state of mode signal 79, upon fire signal 72 being raised on fire pad 70, rather than data stored in the array of memory elements 50-1 to 50-*n* being shifted to the fluid actuators and configuration functions, the data is shifted to the primitive memory arrays of their respective primitives (e.g. M(1) to M(x), M(1) to M(y), and M(1) to M(z)) and to the configuration memory, CM, of the respective group of configuration functions 38-1 to 38-*n*.

FIG. 6 is a block and schematic diagram generally illustrating print component 30 of FIG. 5, where, in lieu of fluidic dies 37-1 to 37-*n* sharing a common intermittent clock signal 35, each fluidic die 37-1 to 37-*n* receives its own corresponding intermittent clock signal, illustrated as clock signals 35-1 to 35-*n* via corresponding clock pads 34-1 to 34-*n*. With reference to FIGS. 2-4, since intermittent clock signals 35-1 to 35-*n* may be separately controlled (e.g., may start and/or stop at differing times), data segments 33-1 to 33-*n* do not need to be of a same length and, thus, may not include filler bit segments 110. Referring to FIG. 6, upon completion of loading of fire pulse groups 100-1 to 100-*n* of data segments 33-1 to 33-*n* into the array of memory elements 50-1 to 50-*n* of the corresponding fluid die 37-1 to 37-*n*, fire signal 72 may be raised to initiate operations on the fire pulse group data (as described above).

FIG. 7 is a block diagram illustrating one example of a fluid ejection system 200. Fluid ejection system 200 includes a fluid ejection assembly, such as printhead assembly 204, and a fluid supply assembly, such as ink supply assembly 216. In the illustrated example, fluid ejection system 200 also includes a service station assembly 208, a carriage assembly 222, a print media transport assembly 226, and an electronic controller 230. While the following description provides examples of systems and assemblies for fluid handling with regard to ink, the disclosed systems and assemblies are also applicable to the handling of fluids other than ink.

Printhead assembly 204 includes at least one printhead 212 which ejects drops of ink or fluid through a plurality of orifices or nozzles 214, where printhead 212 may be implemented, in one example, as print component 30 with fluid actuators (FAs) of actuator groups 36-1 to 36-*n* implemented as nozzles 214, as previously described herein by FIG. 2, for instance. In one example, the drops are directed toward a medium, such as print media 232, so as to print onto print media 232. In one example, print media 232 includes any type of suitable sheet material, such as paper, card stock, transparencies, Mylar, fabric, and the like. In another example, print media 232 includes media for three-dimensional (3D) printing, such as a powder bed, or media for bioprinting and/or drug discovery testing, such as a reservoir or container. In one example, nozzles 214 are arranged in at least one column or array such that properly sequenced ejection of ink from nozzles 214 causes characters, symbols, and/or other graphics or images to be printed upon print media 232 as printhead assembly 204 and print media 232 are moved relative to each other.

Ink supply assembly 216 supplies ink to printhead assembly 204 and includes a reservoir 218 for storing ink. As such, in one example, ink flows from reservoir 218 to printhead assembly 204. In one example, printhead assembly 204 and ink supply assembly 216 are housed together in an inkjet or fluid-jet print cartridge or pen. In another example, ink supply assembly 216 is separate from printhead assembly 204 and supplies ink to printhead assembly 204 through an interface connection 220, such as a supply tube and/or valve.

Carriage assembly 222 positions printhead assembly 204 relative to print media transport assembly 226, and print media transport assembly 226 positions print media 232 relative to printhead assembly 204. Thus, a print zone 234 is defined adjacent to nozzles 214 in an area between printhead assembly 204 and print media 232. In one example, printhead assembly 204 is a scanning type printhead assembly such that carriage assembly 222 moves printhead assembly 204 relative to print media transport assembly 226. In another example, printhead assembly 204 is a non-scanning type printhead assembly such that carriage assembly 222 fixes printhead assembly 204 at a prescribed position relative to print media transport assembly 226.

Service station assembly 208 provides for spitting, wiping, capping, and/or priming of printhead assembly 204 to maintain the functionality of printhead assembly 204 and, more specifically, nozzles 214. For example, service station assembly 208 may include a rubber blade or wiper which is periodically passed over printhead assembly 204 to wipe and clean nozzles 214 of excess ink. In addition, service station assembly 208 may include a cap that covers printhead assembly 204 to protect nozzles 214 from drying out during periods of non-use. In addition, service station assembly **208** may include a spittoon into which printhead assembly **204** ejects ink during spits to ensure that reservoir **218** maintains an appropriate level of pressure and fluidity, and to ensure that nozzles **214** do not clog or weep. Functions of service station assembly **208** may include relative motion between service station assembly **208** and printhead assembly **204**.

Electronic controller **230** communicates with printhead assembly **204** through a communication path **206**, service station assembly **208** through a communication path **210**, carriage assembly **222** through a communication path **224**, and print media transport assembly **226** through a communication path **228**. In one example, when printhead assembly **204** is mounted in carriage assembly **222**, electronic controller **230** and printhead assembly **204** may communicate via carriage assembly **222** through a communication path **202**. Electronic controller **230** may also communicate with ink supply assembly **216** such that, in one implementation, a new (or used) ink supply may be detected.

Electronic controller **230** receives data **236** from a host system, such as a computer, and may include memory for temporarily storing data **236**. Data **236** may be sent to fluid ejection system **200** along an electronic, infrared, optical or other information transfer path. Data **236** represent, for example, a document and/or file to be printed. As such, data **236** form a print job for fluid ejection system **200** and includes at least one print job command and/or command parameter.

In one example, electronic controller **230** provides control of printhead assembly **204** including timing control for ejection of ink drops from nozzles **214**. As such, electronic controller **230** defines a pattern of ejected ink drops which form characters, symbols, and/or other graphics or images on print media **232**. Timing control and, therefore, the pattern of ejected ink drops, is determined by the print job commands and/or command parameters. In one example, logic and drive circuitry forming a portion of electronic controller **230** is located on printhead assembly **204**. In another example, logic and drive circuitry forming a portion of electronic controller **230** is located off printhead assembly **204**. In another example, logic and drive circuitry forming a portion of electronic controller **230** is located off printhead assembly **204**. In one example, data segments **33-1** to **33-*n***, intermittent clock signal **35**, fire signal **72**, and mode signal **79** may be provided to print component **30** by electronic controller **230**, where electronic controller **230** may be remote from print component **30**.

FIG. **8** is a flow diagram illustrating a method **300** of operating a print component, such as print component **30** of FIGS. **2-4**, in accordance with one example of the present disclosure. At **302**, method **300** includes receiving data segments on a number of data pads, such as receiving data segments **33-1** to **33-*n*** on data pads **32-1** to **32-*n*** as illustrated by FIG. **2**, where each data segment comprises a number of segment bits, the number of segment bits including a fire pulse group comprising a number of fire pulse group bits, with the number of segment bits being at least equal to the number of fire pulse group bits, such as illustrated by FIG. **4**A where each data segment **33-1** to **33-*n*** respectively includes a fire pulse group **100-1** to **100-*n***.

At **304**, method **300** includes receiving an intermittent clock signal on a clock pad, such as print component **30** of FIG. **2** receiving an intermittent clock signal **35** on clock pad **34**. At **306**, method **300** includes arranging a number of fluid actuators to form a number of fluid actuator arrays, each array of fluid actuators having a corresponding array of memory elements corresponding to a different one of the data pads, such as actuator groups **36-1** to **36-*n*** of FIG. **2** respectively including an array of fluid actuators **40-1** to **40-*n***, with the arrays of fluid actuators **40-1** to **40-*n*** respectively having a corresponding array of memory elements **50-1** to **50-*n***, with the array of memory elements **50-1** to **50-*n*** respectively having corresponding data pads **32-1** to **32-*n***.

At **308**, method **100** includes serially loading a data segment from the corresponding data pad into each array of memory elements each time the intermittent clock signal is present on the clock pad to store at least the fire pulse group bits, such as respectively loading data segments **33-1** to **33-*n*** (as illustrated by FIGS. **4**A and **4**B) into arrays of memory elements **50-1** to **50-1** so as to respectively store at least fire pulse segments **100-1** to **100-*n***.

Although specific examples have been illustrated and described herein, a variety of alternate and/or equivalent implementations may be substituted for the specific examples shown and described without departing from the scope of the present disclosure. This application is intended to cover any adaptations or variations of the specific examples discussed herein. Therefore, it is intended that this disclosure be limited only by the claims and the equivalents thereof.

The invention claimed is:

1. A set of print components, comprising:

a plurality of dies, each corresponding to a different liquid type, wherein the plurality of dies share an intermittent clock signal and each die includes:

a data pad to receive data segments, each data segment comprising a number of segment bits, the number of segment bits including a fire pulse group comprising a number of fire pulse group bits different for each of the plurality of dies, wherein the number of segment bits is at least equal to the number of fire pulse group bits, and the data segments received on the data pad include a group of filler bits so that the number of segment bits of data segments received by one data pad is the same as a number of segment bits of data segments received by another data pad;

a fluidic actuator array corresponding to the intermittent clock signal and to the data pad, having a corresponding array of memory elements; and

circuitry, connected to the data pad and the fluidic actuator array, and including the corresponding array of memory elements, the circuitry configured such that a rising edge of the intermittent clock signal causes the corresponding array of memory elements to serially receive a data segment from the data pad and to store at least the fire pulse group bits.

2. The set of print components of claim **1**, each fluidic actuator array having a corresponding group of configuration functions.

3. The set of print components of claim **2**, each array of memory elements including a first portion of memory elements corresponding to the group of configuration functions and a second portion of memory elements corresponding to the fluidic actuator array.

4. The set of print components of claim **1**, the array of memory elements comprising a chain of memory elements adapted to function as a serial-to-parallel data converter.

5. The set of print components of claim **4**, wherein the array of memory elements includes a sequential logic circuit.

6. The set of print components of claim **5**, the sequential logic circuit adapted to function as a serial-in, parallel-out shift register.

7. The set of print components of claim **1**, wherein the corresponding array of memory elements is to load the data segments from the corresponding one of the plurality of data pads only when the intermittent clock signal is present on the clock pad.

8. The set of print components of claim **1**, wherein the fire pulse group does not include a start bit sequence that identifies a beginning of the fire pulse group.

9. The set of print components of claim **8**, wherein the print component does not include data parser circuitry to detect the beginning of the fire pulse group.

10. A set of print components, comprising:
- a plurality of dies, each corresponding to a different liquid type, wherein the plurality of dies share an intermittent clock signal and a fire signal, and each die includes:
  - a fire pad to receive the fire signal;
  - a clock pad to provide the intermittent clock signal provided at differing times;
  - at least one data pad to receive data segments, each data segment comprising a number of segment bits, the number of segment bits including a fire pulse group comprising a number of fire pulse bits different for each of the plurality of dies, wherein the number of segment bits is at least equal to the number of fire pulse group bits, and the data segments received on the at least one data pad include a group of filler bits so that the number of segment bits of data segments received by one data pad is the same as a number of segment bits of data segments received by another data pad;
  - a fluidic actuator array corresponding to the intermittent clock signal and to the at least one data pad, having a corresponding array of memory elements; and
  - circuitry, connected to the at least data pad and the fluidic actuator array, and including the corresponding array of memory elements, the circuitry configured such that a rising edge of the intermittent clock signal causes the corresponding array of memory elements to serially receive a data segment from the at least one data pad and to store the fire pulse bits.

11. The set of print components of claim **10**, the array of memory elements comprising a chain of memory elements adapted to function as a serial-to-parallel data converter.

12. The set of print components of claim **11**, the array of memory elements comprising a sequential logic circuit.

13. The set of print components of claim **12**, the sequential logic circuit adapted to function as a serial-in, parallel-out shift register.

14. The set of print components of claim **10**, including:
- a plurality of clock pads, wherein each clock pad is to receive a corresponding one of a plurality of intermittent clock signals that include the intermittent clock signal, and corresponds to each of the plurality of dies.

15. The set of print components of claim **14**, wherein the plurality of intermittent clock signals are provided at differing times.

16. The set of print components of claim **10**, wherein the corresponding array of memory elements is to load the data segments from the corresponding one of the plurality of data pads only when the intermittent clock signal is present on the clock pad.

17. The set of print components of claim **16**, wherein the corresponding array of memory elements is to serially load the data segments in response to the clock pad receiving a rising edge of the intermittent clock signal.

18. The set of print components of claim **10**, wherein the fire pulse group does not include a start bit sequence that identifies a beginning of the fire pulse group.

19. The set of print components of claim **18**, wherein the print component does not include data parser circuitry to detect the beginning of the fire pulse group.

20. A plurality of print components, comprising:
- a fluidic actuator array of each component, each array corresponding to a different liquid type, wherein the plurality of fluidic actuator arrays share an intermittent clock signal and a fire signal, and each component includes:
  - a clock pad to provide the intermittent clock signal;
  - a fire pad to receive the fire signal;
  - at least one data pad to receive data segments, each data segment comprising a number of segment bits, the number of segment bits including a fire pulse group comprising a number of fire pulse group bits, wherein the number of segment bits is at least equal to the number of fire pulse group bits, and the data segments received on the at least one data pad include a group of filler bits so that the number of segment bits of data segments received by one data pad is the same as a number of segment bits of data segments received by another data pad; and
  - circuitry, connected to the at least one data pad and the fluidic actuator array, and including at least one corresponding memory element, wherein a rising edge of the intermittent clock signal causes the at least one corresponding memory element to serially receive a data segment from the at least one data pad and to store the at least one fire pulse group bit.

21. The plurality of print components of claim **20**, the at least one corresponding memory element comprising a chain of memory elements adapted to function as a serial-to-parallel data converter.

22. The plurality of print components of claim **21**, the at least one corresponding memory element comprising a sequential logic circuit.

23. The plurality of print components of claim **22**, the sequential logic circuit adapted to function as a serial-in, parallel-out shift register.

24. The plurality of print components of claim **20**, comprising a plurality of data pads, each of which corresponds to one of the plurality of fluidic actuator arrays.

25. The plurality of print components of claim **20**, the at least one fluidic actuator array arranged to form a plurality of primitives, each primitive having a same number of fluid actuators, and the at least one corresponding memory element including a first portion corresponding to at least one configuration function and a second portion corresponding to the at least one fluidic actuator array.

26. The plurality of print components of claim **20**, the print component comprising a printhead.

27. The plurality of print components of claim **20**, the print component comprising a mode pad to receive a mode signal, wherein a state of the mode signal indicates whether a data value stored in the at least one corresponding memory element corresponds to the at least one fluidic actuator array.

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